COURSE CODE	COURSE TITLE	L	T	P	C
UIT2702	CLOUD AND DISTRIBUTED COMPUTING	2	0	2	3

OBJECTIVES:

The objective of this course is to enable the students to

- Explain about distributed system and cloud models.
- Apply distributed computational model and understand the need for cloud computing.
- Learnabout cloud recovery and storage.

UNIT I INDRODUCTION TO DISTRIBUTED SYSTEM 6

Characterization of Distributed System: Introduction-Examples of distributed system, Trends in distributed system, focus on resource sharing, Challenges – System models – Inter Process Communication – Remote Invocation – Indirect Communication – Case study: World Wide Web.

UNIT II DISTRIBUTED ALGORITHMS 6

Message Passing, Leader Election, Distributed Models, Causality and Logical Time, Global State & Snapshot and Distributed Mutual Exclusion- Token based approaches, Consensus & Agreement, Checkpointing & Rollback Recovery - Introduction classical distributed algorithms - Algorithm for Recording Global State and Snapshot - Time and Clock Synchronization in Cloud Data Center - Key Challenges.

UNIT III CLOUD VIRTUALIZATION 6

Features of Today's Cloud-Layered Cloud Architecture Design – NIST Cloud Computing Reference Architecture: public, private, hybrid. Service provider interfaces: SaaS, PaaS, IaaS – Virtualization Technology - Hardware Independence, Server Consolidation, Resource Replication, Operating System-Based Virtualization, Hardware-Based Virtualization, Virtualization management and considerations – Use case and Example.

UNIT IV CLOUD STORAGE AND RECOVERY 6

Fundamental cloud architectures - workload distribution, resource pooling, dynamic scalability, elastic resource capacity, service load balancing, cloud bursting, elastic disk provisioning, redundant storage, Case Study Example - Advanced Cloud Architectures, Hypervisor clustering, Load balanced virtual server instances, Non-Disruptive service relocation, Zero-downtime, Cloud balancing, Resource reservation, Dynamic failure detection and recovery, Bare-metal provisioning, Rapid provisioning, Storage workload management, Case Study Example.

UNIT V CLOUD PLATFORMS AND APPLICATIONS 6

Cloud platform: Microsoft Azure, Amazon Web Services, Google Cloud, IBM Cloud and CloudLinux– APIs - Microservices - Docker - Kubernetes - Applications and Use Cases.

Lectures: 30 HOURS

LAB SESSIONS:

1. Inter-process communication
2. Remote Method Invocation
3. Message passing and distributed mutual exclusion
4. Clock and Time synchronization
5. Virtual instances and server consolidation with fault-tolerance
6. Task scheduling and load balancing
7. Hypervisor clustering
8. Hosting microservices on cloud platforms
9. Dockerization and Kubernetes
10. Mini project

Lab: 30 HOURS

TOTAL: 60 HOURS

TEXTBOOK:

1. Rajiv Misra, Yashwant Singh Patel, Cloud and Distributed Computing: Algorithms and Systems, Wiley India.

2. Coulouris George, Dollimore Jean, Kindberg Tim, Blair Gordon, Distributed Systems Concepts and Design, Pearson Education, 5th Edidtion

REFERENCES:

1. Danielle Ruest, Nelson Ruest, Virtualization: A Beginner's Guide, McGraw-Hill Osborne Media, 2009.
2. Thomas Erl, Ricardo Puttini, Zaigham Mahmood, Cloud Computing: Concepts, Technology & Architecture, First Edition, Pearson Education 2013.
3. John W.Rittinghouse and James F.Ransome, Cloud Computing: Implementation, Management, and Security, CRC Press, 2010.
4. Michael J.Kavis, "Architecting the Cloud". Wiley India, 2014.

COURSE OUTCOMES:

On successful completion of this course, the student should be able to

1. Understand how distributed system concepts apply inside cloud.
2. Apply the distributed algorithms for effective scheduling.
3. Analyze the need for virtualization and apply it in a cloud environment.
4. Understand the architecture, infrastructure, and delivery models of cloud computing.
5. Explore cloud tools and build applications.

COs	POs												PSOs	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	2	2	1											
CO2	2	2		1	2									2
CO3		2	1	1	1								1	2
CO4		2												2
CO5		2	1	1	2							1	1	2